

Problem Statements - Assessment 1

Question 1 – Water Jug Problem

The Water Jug Problem is a classic Artificial Intelligence state-space search problem. The objective is to obtain exactly 2 gallons of water in a 4-gallon jug using only two unmarked jugs of capacities 4 gallons and 3 gallons. The allowed operations include filling a jug, emptying a jug, and pouring water from one jug to another. Since there are no measuring marks on the jugs, the solution must be obtained by exploring different states systematically. This problem demonstrates how AI search techniques can be used to find an optimal sequence of actions to achieve a desired goal.

Question 2 – Mars Rover Intelligent Agent

A Mars Rover is an autonomous intelligent agent designed to explore the surface of Mars without continuous human intervention. It must collect information from its surroundings using sensors, analyze rocks and soil, avoid obstacles, and send scientific data back to Earth. Since the Martian environment is uncertain and constantly changing, the rover must make intelligent decisions based on the information it perceives. This problem focuses on identifying the rover's percepts, actions, operating environment, performance measures, and the most suitable AI agent architecture that enables autonomous exploration.

Question 3 – 8 Queens Problem

The 8-Queens Problem is a well-known Constraint Satisfaction Problem in Artificial Intelligence. The objective is to place eight queens on an 8×8 chessboard such that no queen attacks any other queen. A queen can attack horizontally, vertically, and diagonally, making the placement challenging. The problem requires exploring different board configurations while eliminating invalid placements using backtracking. It demonstrates how AI search algorithms efficiently solve complex combinatorial problems by reducing unnecessary searches.

Question 4 – OLA Cab Booking

The OLA Cab Booking System is an example of a goal-based problem-solving agent in Artificial Intelligence. A customer wants to travel from one location to another by selecting a suitable cab based on factors such as fare, comfort, availability, and waiting time. The system must evaluate multiple cab options and choose the most appropriate one to satisfy the user's goal. This problem illustrates how intelligent agents make decisions based on user preferences and available information to provide an efficient and reliable transportation service.

Question 5 – Uniform Cost Search

A logistics company operates a network of warehouses connected by routes with different travel costs. The objective is to determine the least-cost path from the starting warehouse to the destination warehouse while minimizing transportation expenses. Since different routes have different costs, the algorithm must always explore the path with the lowest cumulative cost before considering more expensive alternatives. This problem demonstrates the application of the Uniform Cost Search algorithm, which guarantees finding the optimal path when all edge costs are positive.